

Manufacturing ERP + MES — Fresh All-in-One Product & Technical Plan

Target: small/medium Indian factories currently using Excel, paper, registers and WhatsApp. ERP is the first product; MES is a separate application and is integrated after ERP is proven with a real customer.

Metric	Value
Total backlog items	114
ERP items	90
MES items	22
Platform items	2
Estimated hours	2224
Estimated person-days (8h)	278.0
Dedicated DB model	One PostgreSQL database per customer + central control DB
ORM	None — Repository Pattern + parameterized SQL + Stored Procedures
Initial hosting	Docker + Railway; migration-ready for Azure

Architecture

Area	Decision / Standard
Frontend	React + TypeScript + Material UI + AG Grid
Backend	Node.js + TypeScript REST API + OpenAPI
Design	SOLID + Controller -> Service -> Repository -> Database
ORM	None
Data access	Repository Pattern; parameterized SQL and Stored Procedures
Database	PostgreSQL
Tenant isolation	One dedicated PostgreSQL database per customer
Control DB	Tenants, subscriptions, plans, features, DB registry, identity metadata only
Business data	Stored only in customer-specific DB
Tenant resolution	Server-side from authenticated identity/membership; never from arbitrary browser DB ID
Files	Azure Blob Storage with tenant-scoped paths and authorization
Async	Redis + Node.js worker for large exports/background jobs
Deployment	Docker; Railway initially; architecture migration-ready for Azure
CI/CD	GitHub Actions
Security	JWT/refresh tokens, RBAC, validation, parameterized SQL, rate limiting, audit, HTTPS, secrets management
Reports	Server-side PDF generation; authenticated download and audit
Exports	XLSX/CSV export, server-side permission checks, async large exports
ERP/MES	Separate applications; no shared DB tables
Integration	Versioned REST APIs first; events/outbox-ready later
MES initial scope	Manual shop-floor execution first; no PLC/IoT dependency
Future	MQTT/OPC-UA, advanced analytics and AI after real operational data exists

PDF, Excel and CSV Export Requirements

Feature	PDF	Excel/XLSX	CSV	Security / Technical
Inventory report	Yes	Yes	Yes	Tenant DB; view/export permission; audit
Purchase order	Yes	Yes	Yes	Authorized users only; PDF template
GRN	Yes	Yes	Yes	Document audit; tenant-scoped file
Sales order	Yes	Yes	Yes	Customer data protected
Delivery/dispatch	Yes	Yes	Yes	Inventory consistency
Production order	Yes	Yes	Yes	ERP/MES traceability
Production result	Yes	Yes	Yes	MES integration audit
Quality/NCR	Yes	Yes	Yes	Sensitive operational data
OEE/downtime	Yes	Yes	Yes	MES permissions and filters
Large report	Async	Async	Async	Redis/worker + secure temporary download
File retention	Configurable	Configurable	Configurable	Tenant-specific retention and cleanup

Jira Backlog Summary

Epic	Product	Items	Hours
Product Discovery & Pilot	ERP	1	12
Platform & Architecture Foundation	ERP	9	100
Customer DB Provisioning & Isolation	ERP	16	272
Authentication & RBAC	ERP	3	60
Master Data	ERP	3	60
BOM & Routing	ERP	2	44
Procurement	ERP	3	70
Inventory & Traceability	ERP	5	124
Sales & Dispatch	ERP	2	48
Production Planning	ERP	4	84
Basic Quality	ERP	3	66
Manufacturing Costing	ERP	2	48
Workflow & Customization	ERP	3	72
Dashboards & Reporting	ERP	2	48
PDF Reports & Documents	ERP	5	96
Excel/CSV Export & Data Export	ERP	6	98
SaaS Administration & Subscription	ERP	2	44
Security, Audit & Compliance	ERP	5	96
Files, Storage & Backup	ERP	2	36
Deployment, CI/CD & Observability	ERP	10	162
ERP Customer Pilot & Hardening	ERP	2	56
MES Foundation	MES	1	24
MES Identity & Operator	MES	1	20
MES Work Centers, Machines & Shifts	MES	2	36
MES Production Execution	MES	3	72
MES Quality & Traceability	MES	2	52

Epic	Product	Items	Hours
MES Downtime & OEE	MES	2	52
MES Shop-Floor UI	MES	2	48
ERP-MES API Integration	MES	5	124
MES Data Export & Reports	MES	2	40
IoT/Machine Gateway Future	MES	2	36
Future Analytics & AI	Platform	2	24

Key Implementation Rules

1. Customer business data lives in a dedicated PostgreSQL database; never use a shared business database with `tenant_id` as the primary isolation mechanism.
2. Central control DB contains platform metadata only; the browser never receives database credentials or connection strings.
3. Use Controller → Service → Repository Interface → Repository Implementation → Stored Procedure/Parameterized SQL.
4. No ORM. Complex writes use stored procedures with explicit transaction boundaries; simple reads may use parameterized SQL.
5. PDF, Excel and CSV exports are tenant-aware, permission-checked and audited. Large exports run asynchronously through a worker.
6. ERP and MES are separate applications and databases. Integrate through versioned REST APIs; do not share database tables.
7. Start MES with manual operator execution. Defer PLC/IoT/MQTT/OPC-UA until the core ERP/MES workflows are proven.
8. Avoid customer-specific code forks. Use configuration, workflows, feature flags and custom fields.
9. Build security, backup/restore, monitoring and isolation testing as first-class backlog items, not post-launch tasks.